Version 1 Review

Overview from my project specs:

The Canvas Backup Tool will allow users, primarily instructors and instructional designers, to extract their downloaded IMS Common Cartridge file (.imscc) of their Canvas course into logically organized and named folders based on their course's module structure, so they can easily find and retrieve content outside of the Canvas environment.

The Canvas Backup Tool will be a desktop application (using Tkinter). Users will:

- Download their course IMS Common Cartridge file
- Open the Canvas Backup Tool application to find and select their .imscc file.
- Wait for the program to unzip the .imscc file, parse the xml, create a folder structure in the same folder as their .imscc file, and copies files into the logical folder structure that emulates the Canvas module setup as mapped in the xml.

Progress so far:

The desktop app/GUI will be implemented in version 2, but for now this is a functional script that has hit the major milestones listed above:

- Extracts .imscc package (into a temp folder)
- Parses imsmanifest.xml to index resource IDs to file paths and extract module hierarchy and item order
- Creates folder structures based on modules with clean titles and copies matching raw files
- ADDED: Automatically generates a log report for successful and failed file items
- ADDED: Deletes temp folder after process is complete

In detail:

Video to Canvas Studio walkthrough.

This shows the script successfully extracts, creates and cleans up the temp folder, and creates the course back up folder with module folders and copies over the files. I also show the summary log that reports on successful and missing items.

Issues:

The biggest issue is that I assumed quizzes and discussions would look like assignments (HTML) but they do not and they are associated with multiple other XML files in the

imsmanifest. The way that I handled this issue is by generating a summary log to let the user know what was missing. I don't think missing discussions is a big deal because if Canvas is truly down, then you can't really do much for online discussions. However, I do think backing up quizzes/tests would be important.

Here is an example of Module 1 in the imsmanifest.xml file:

```
<organizations>
  <organization identifier="org_1" structure="rooted-hierarchy">
    <item identifier="LearningModules">
      <item identifier="g0973f73bfc3c4b3620eb9f64d8cbfa8b">
        <title>Chapter 1: An Introduction to Sociology</title>
        <item identifier="g28b2f4514a19356951673fcf7810d438" identifierref="g440383174ad686484ee55ad49173269a">
          <title>Chapter 1 Reading</title>
        </item>
        <item identifier="gfd99252ece154b13c299361589bec0d9" identifierref="g8fbbe296db1355daeef42d214c1fb581">
          <title>Chapter 1 Slides</title>
        </item>
        <item identifier="gc310d998757bd367e1c5a6982121d6bb" identifierref="g86de75b25b1adc19b9afad2241a8b5d3">
          <title>Chapter 1 Quiz</title>
        </item>
        <item identifier="gbc838f7ff6afa457145aa16b31666873" identifierref="g4bfae4b575b59d0d2c5757723e735318">
          <title>M1 Discussion</title>
        </item>
      </item>
```

I can extract the reading and slides, but not the quiz and discussion. I can ID all 4 of these items. In the screenshot above, I've copied the ID reference for Chapter 1 Quiz and here's where else it exists:

```
<resource identifier="g86de75b25b1adc19b9afad2241a8b5d3" type="imsqti_xmlv1p2/imscc_xmlv1p1/assessment">
  <file href="g86de75b25b1adc19b9afad2241a8b5d3/assessment_qti.xml"/>
  <dependency identifierref="i2a3b2fb4b167e228bf5358b9250f91ea"/>
</resource>
<resource identifier="i2a3b2fb4b167e228bf5358b9250f91ea" type="associatedcontent/imscc_xmlv1p1/learning-
application-resource" href="g86de75b25b1adc19b9afad2241a8b5d3/assessment_meta.xml">
  <file href="g86de75b25b1adc19b9afad2241a8b5d3/assessment_meta.xml"/>
  <file href="non_cc_assessments/g86de75b25b1adc19b9afad2241a8b5d3.xml.qti"/>
</resource>
```

So this one quiz is linked to these different xml files. Does that mean I would need to parse each of these xml files to copy it over? Do I even want to do that? The quizzes are QTI files, so the user would have to find a way to convert them to be usable. I like people to have choices, so I'll list it as a future "nice to have" but I've got these logged for now in the summary log as missing.

Version 2 milestones:

- Build GUI with Tkinter
  - Add file input and destination folder
  - Add message for completion
- Test edge cases

    - Test a variety of courses
- Error handling (probably done alongside testing)
    - Add user alerts/safety checks for issues with files
- Separate out main.py's blocks into their own modules
    - Extraction, parsing, building modules, etc.
- Update documentation
    - Write user instructions/update README

Self-Reflection

I am pretty satisfied with the progress. I do think I can finish the project soon if I don't pursue copying over quizzes. Directory work with os and shutil was pretty straightforward, but navigating XML namespaces and non-file items took some research. One thing I didn't consider in my initial specs was cleaning up the temp folder. I had to keep deleting it during testing and realized this should probably be taken care of.